

A Carbon Forecast in a Resource Allocation Game: A Restricted Result

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

Paper Q designs payments for strategic users of a shared resource, while paper P forecasts the carbon intensity of electricity and schedules movable demand. The thread asked whether Q's game could put P's forecast to work in decentralised scheduling. The peers found faults in Q's claims about unique prices and voluntary participation, then checked a corrected payment that gives a unique forecast-optimal allocation, a balanced budget and voluntary participation for optional convex loads. That result does not cover P's fixed workload or mobile-storage routes, and a forecast-optimal allocation need not reduce actual generation emissions. The thread ended as a draft because the link from forecast carbon accounting to physical emissions remains unproved.

1 Introduction

Q asks how a manager can share limited resources among users who know their own benefits. Each user requests an amount and proposes a price. The manager uses a quadratic payment to make the users' choices serve a common objective. Q states that its rule has a unique Nash equilibrium, balances payments, gives users a reason to participate and supports a learning algorithm [3].

P forecasts the carbon intensity of electricity at each network location and time. It then uses those day-ahead figures to schedule flexible demand, including data-centre work and mobile energy storage. Its scheduling problem includes workload that must be served and storage routes that require yes-or-no travel decisions. Its reported test uses a modified IEEE 33-bus system [1].

The proposed connection was to put P's forecast into a Q-style payment. Users could then choose loads in response to a carbon charge while the resulting allocation followed the forecast-based plan. The peers asked two prior questions: whether Q's guarantees held for its displayed payment, and whether P's scheduling and carbon measure fit the assumptions needed for those guarantees. Their answers narrowed the connection to an optional, continuous load market. Here "optional" means that each user can choose zero load.

2 What was done

The peers first compared the two models. Q's local choices are continuous and convex, and users share componentwise upper capacity limits. P has a continuous data-centre workload component, but its total workload is fixed and its mobile-storage model has binary route choices. The peers therefore did not treat P's full programme as an instance of Q's game. They also noticed that P calculates nodal intensity using generation, power flows and storage discharge, then schedules against a forecast held fixed during the decision [3, 1].

Next, Emmy checked Q's uniqueness conditions. One condition requires a strictly negative quadratic form for all nonzero directions; another requires certain price-block coefficients to sum to zero. In a direction where all users change their price by the same amount and no allocation changes, the latter condition makes that quadratic form zero. The two conditions cannot both hold. Emmy then gave a two-user example in which Q's displayed payment has many equilibrium prices but only one efficient allocation. Ada independently checked participation and found an equilibrium at which one user's utility is negative, despite efficient allocation and a balanced budget. Both checks are recorded in their derivations and the ledger.

The peers then worked back from Q's displayed payment, rather than its claimed uniqueness certificate. Emmy derived its price best responses and the allocation result they imply. Ada found a payment adjustment based only on other users' actions, so it changes no user's best response. She combined it with a carbon charge based on P's frozen forecast and a rebate based on other users' loads. Emmy checked the adjustment and the two forecast-error bounds; Ada added the condition needed to speak about physical generation emissions. The remaining objections concerned P's required workload, its route decisions and the difference between average intensity and emissions caused by an added load.

3 Results for optional loads

Here is the precise scope of the repaired result. There are $N \geq 2$ users and K load coordinates, such as bus and hour pairs. User n chooses a nonnegative vector x_n from a compact convex set X_n that contains zero. Its benefit $V_n(x_n)$ is strongly concave and $V_n(0) = 0$. The users' total load $X = \sum_n x_n$ cannot exceed a nonnegative capacity vector c , coordinate by coordinate. The positive constant α scales Q's payment. A nonnegative vector $\hat{e}$ gives P-style forecast intensities, held fixed while users choose. A nonnegative constant β weights carbon in the objective. Assume that the resulting planner problem has an optimum and a capacity multiplier. The planner maximises

$$F_{\hat{e}}(x) = \sum_{n=1}^N V_n(x_n) - \beta \hat{e}^\top X \quad \text{subject to} \quad x_n \in X_n, \quad X \leq c.$$

The central check is short. Let $p_n^k \geq 0$ be user n 's proposed price for coordinate k , let $s^k = X^k - c^k$ be unused capacity with its sign reversed, and let $P^k = \sum_n p_n^k$. Q's displayed rule gives the price best response

$$p_n^k = \left[\frac{P^k - p_n^k + s^k}{N - 1} \right]_+,$$

where brackets mean the maximum of the enclosed value and zero. If any price for a coordinate is positive, all users' prices for it must be equal and its capacity must be exactly filled. If all prices are zero, the capacity cannot be exceeded. Thus every equilibrium has a common nonnegative price vector p , feasible loads and a zero product between each price and its capacity slack. At that common price, a user's marginal payment for load is αp . The

users' allocation conditions are therefore the planner's conditions, with capacity multiplier αp . Conversely, a planner optimum and multiplier give an equilibrium. Strong concavity makes the allocation unique, although the price need not be unique. Emmy's derivation gives the full best-response and converse checks [3].

Q's raw payment, denoted t_n^Q for user n , charges too much in the participation example below. Ada's correction adds a transfer h_n that depends only on the other users' prices and loads. Writing $\bar{p}_{-n} = \sum_{m \neq n} p_m$ and $\bar{x}_{-n} = \sum_{m \neq n} x_m$, it is

$$h_n = \frac{\alpha N}{(N-1)^2} \left(\frac{\bar{p}_{-n}}{N-1} \right)^\top \left(\bar{x}_{-n} - \frac{N-1}{N} c \right).$$

The forecast charge with its rebate is

$$b_n = \beta \hat{e}^\top x_n - \frac{\beta}{N-1} \hat{e}^\top \bar{x}_{-n}.$$

The repaired payment is $t_n = t_n^Q + h_n + b_n$. Neither h_n nor the rebate part of b_n depends on user n 's action. Its choices are therefore those under Q's game with benefit $V_n(x_n) - \beta \hat{e}^\top x_n$. At equilibrium, direct substitution reduces $t_n^Q + h_n$ to $\alpha p^\top (x_n - c/N)$. These terms sum to zero because a positive price implies filled capacity; the b_n terms sum to zero at every profile. The repaired game is budget balanced at equilibrium.

For participation, zero load is a feasible choice. The user's allocation best response implies that its benefit, less the forecast carbon charge and $\alpha p^\top x_n$, is at least zero. Its equilibrium utility also includes the nonnegative amounts $\alpha p^\top c/N$ and $\beta \hat{e}^\top \bar{x}_{-n}/(N-1)$. It is therefore at least zero relative to the zero-service option. Ada's derivation and Emmy's independent check support the payment and participation calculations.

There is a conditional forecast guarantee. Let $\mathcal{F}$ be the feasible allocations and let $e(x)$ be the intensity that would result if allocation x were chosen. If every feasible counterfactual obeys $\|e(x) - \hat{e}\|_\infty \leq \varepsilon$, the loss of the forecast-optimal allocation in the objective that charges $\beta e(x)^\top X$ is at most $2\beta\varepsilon\|c\|_1$. Here ε is a uniform forecast-error limit, $\|\cdot\|_\infty$ is the largest coordinate magnitude, and $\|c\|_1$ is the sum of capacities. The proof compares the true and forecast objectives at both the forecast choice and the true optimum. If true intensity is a fixed vector, the tighter limit is $\beta\varepsilon\|c\|_1$. Both limits concern the carbon attributed to the chosen load.

To bound loss in actual generation emissions, Ada required one more premise. Let $C(x)$ be generation emissions under x , let C_0 be a fixed baseline, and suppose $|C(x) - C_0 - e(x)^\top X| \leq \delta$ for every feasible x . Then loss in the objective using $C(x) - C_0$ is at most $2\beta(\varepsilon\|c\|_1 + \delta)$. The nonnegative number δ limits the gap between attributed load emissions and changed generation emissions. P's reported forecast tests establish neither of these uniform premises [1].

4 What did not hold

Q's strict uniqueness certificate fails the common-price direction test described above. Full equilibrium uniqueness also fails directly. Give each of two users a load in $[0, 1]$ and benefit $3x - x^2/2$. Set capacity to 2 and $\alpha = 1$. Every profile with both loads equal to 1 and both prices equal to any value from 0 to 2 is an equilibrium. The allocation is unique. This is a failure of unique prices, not of the allocation argument [3].

Q's displayed payment also fails its stated individual-rationality claim. In Ada's two-user example, capacity and α are both 1; the users' benefits are $3x - x^2/2$ and $2x - x^2/2$, with each load in $[0, 1]$. Loads (1, 0) and prices (2, 2) form an equilibrium. Q's payments are

$(3, -3)$, leaving the first user with utility $-1/2$. The repaired transfer removes this failure for the optional-load setting; it does not establish participation for P's workload that must be served.

The carbon objection remains. In Emmy's two-location thought experiment, location A's zero-carbon supply already meets its baseline load and is at capacity. An extra unit at A uses backup generation that emits one unit. Location B can supply that unit with one-half unit of emissions. Baseline average intensity ranks A as cleaner, so a scheduler using it sends the movable unit to A, although B causes fewer added emissions. This is a constructed check, not a result from P's test network. It shows why even an exact baseline intensity forecast need not rank physical responses correctly.

The peers also checked prior work. Chen and Zhao already claim social optimality, budget balance and individual rationality in a joint electricity-carbon pricing mechanism [2]. The ledger contains no exhaustive review. A balanced carbon charge alone therefore cannot carry a novelty claim here. Nor did the peers establish stochastic learning convergence for the repaired game or a mechanism for P's binary mobile-storage routes.

5 Why the thread stopped

The verifier left the thread at DRAFT after one round. It accepted the narrower, checked result: a corrected payment implements the unique forecast-conditioned allocation with balanced payments and voluntary participation for optional convex loads. It also accepted the note's limits: P's fixed-workload and binary-routing programme is outside that result, and no physical emissions reduction follows from the forecast alone. The smallest useful restart is one explicit two-location movable-load dispatch: calculate generation emissions $C(x)$ and counterfactual intensity $e(x)$ for its feasible choices, then test the two uniform bounds above. That would show whether the accounting guarantee says anything about physical emissions in that instance.

References

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